Sign Language Translation Assistant

Architecture Seminar

Brent Garey, Paschal Corrigan, Bhargav Singapuri, and Ashley Wu

Sign Language Recognition

As human interpretable problem:

- Observe a speaker communicating using their hands, upper body, and facial expressions
- Understand the communication through situational context

As a computer vision problem:

- Observe features from sequences of captured images. These are hand and face crops, and pose estimations
- Create functions to map features to embeddings, which have been learned through supervised training (Encoding)
- Create functions to map embeddings to meaning, i.e English text (Decoding)

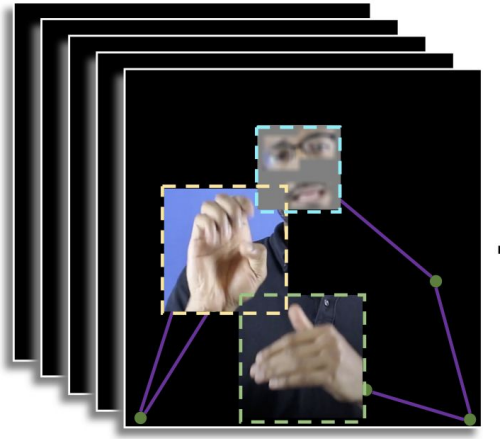


Image: Gueuwou et al. (2025), SHuBERT, [1].

Getting the Features (Inputs)

Dataset

- We will be using the How2Sign dataset for videos of continuous signed sentences. The complete dataset includes 80 hours of video
- We will be using only frontal videos, with green screen backgrounds

Preprocessing:

- We will be processing videos at 15 frames per second
- Key pose points and hand and face crops will be extracted using MediaPipe

Note: The how2 sign frame on the right shows a blurred face for anonymity, we have access the facial features which were given with consent.



Image: How2Sign (Duarte et al., 2021) [2]

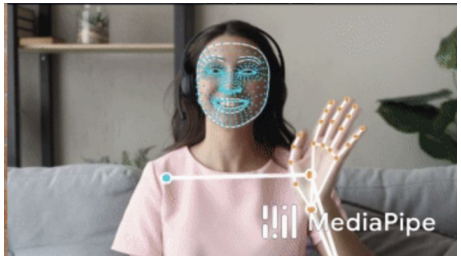
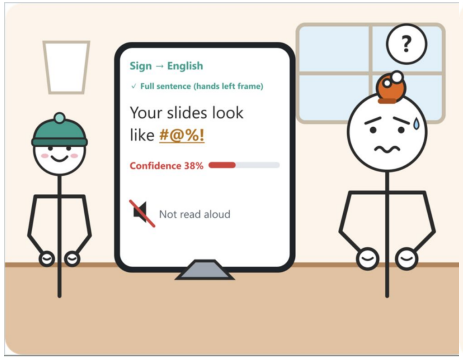
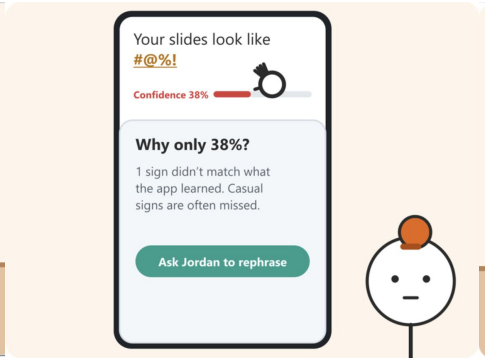


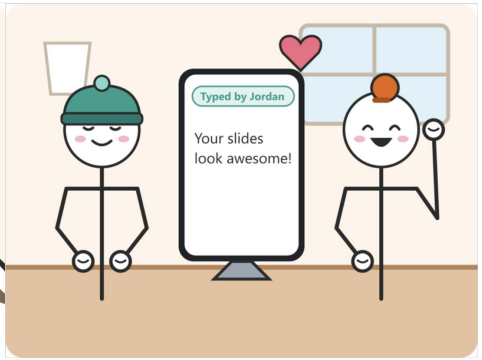
Image: Grishchenko & Bazarevsky (2020), MediaPipe Holistic [3]

Storyboard B and the Translation Feature







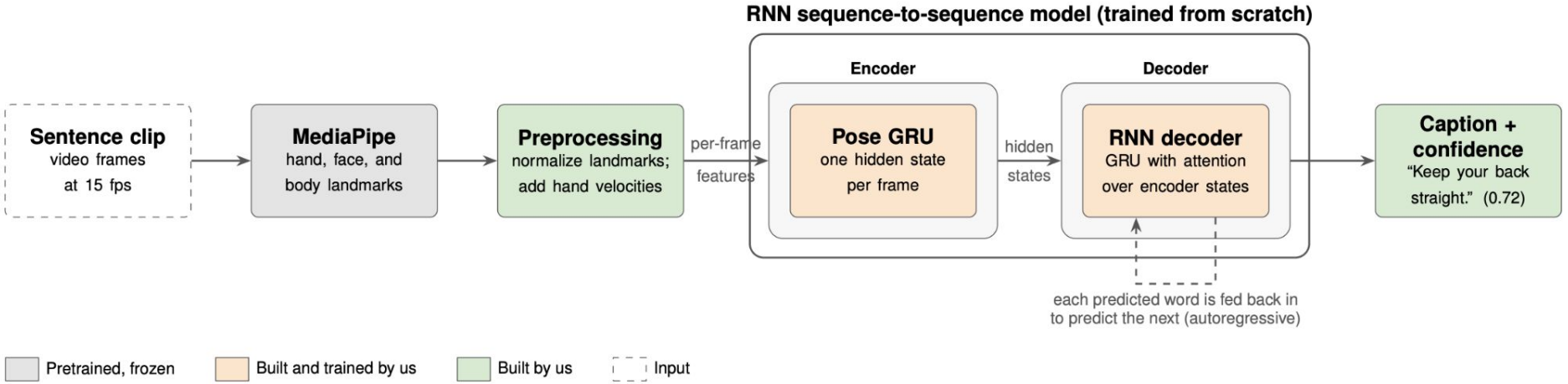


1. Mistranslation **2. Confidence Flag** **3. Retry** **4. Correct/Typed**

1. Inputs are collected from a webcam in real time
2. We generate a caption one token at a time
3. The output token is assigned a probability in the decoding step
4. The average of log probabilities builds the sentence level confidence score (more robust with low-probability tokens)

Architecture I: Classical (Proposal)

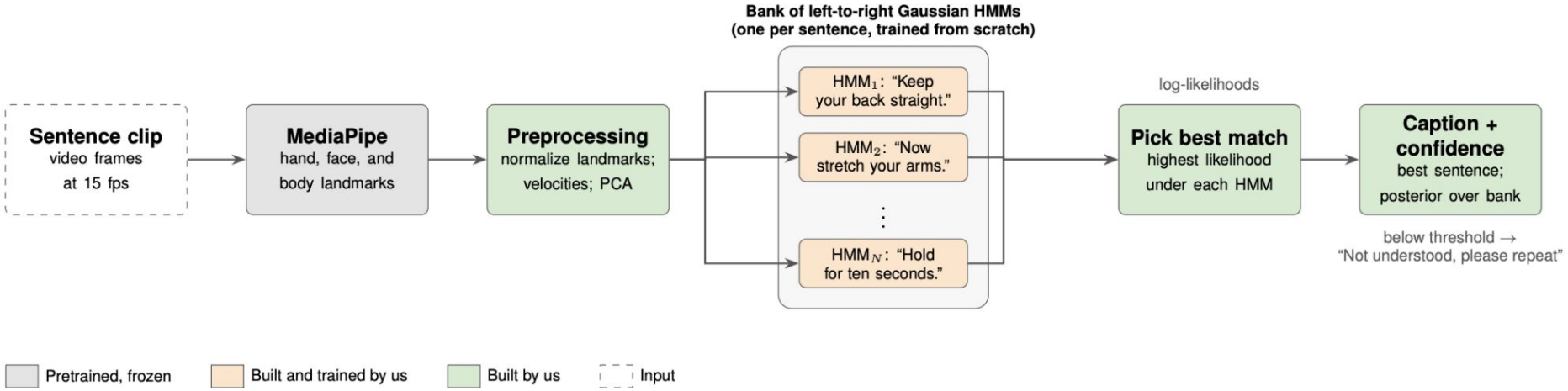
Pose GRU Encoder, RNN Decoder (LSTM or GRU)
- RNNs were the de facto approach to seq2seq problems before transformers
- Exploding/vanishing gradients, require n sequential steps (cannot be parallelized)
- Transformers have been popular for about 10 years and generally exceed RNN performance with large datasets, so we propose this as ‘classical’ in our special case



Architecture I: Classical (Fallback)

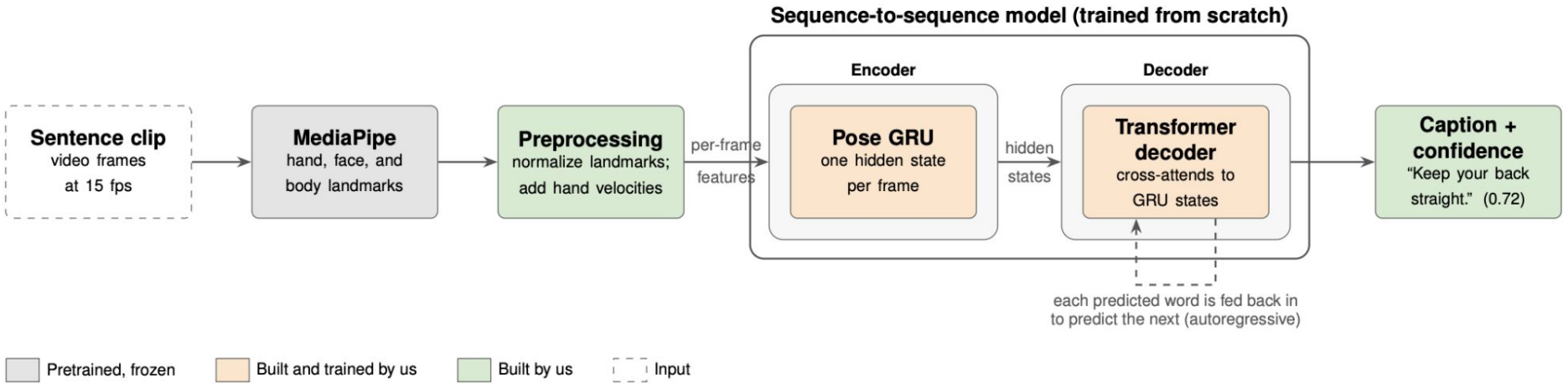
Hidden Markov Model with a Sentence Bank

- Hidden markov models cannot generate freeform English
- For a shot at reasonable performance we could restrict users to a sentence bank
- Corresponds to pre-neural net classical approaches for sign translation [4]



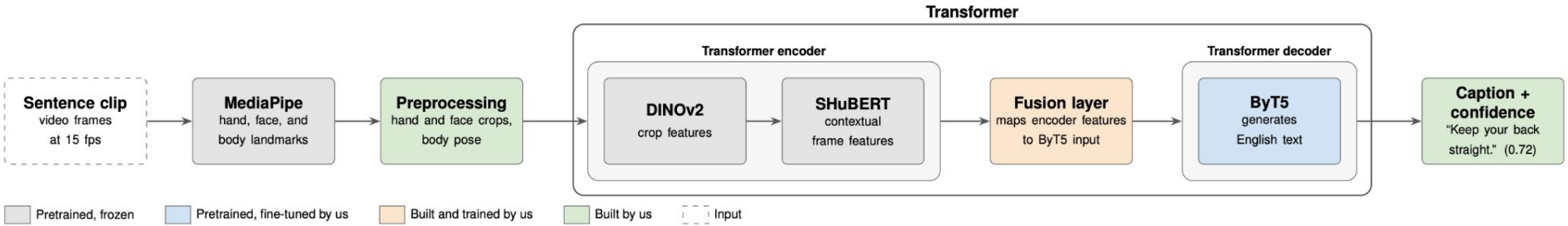
Architecture II: Baseline

Pose GRU Encoder, Transformer Decoder
- Self trained GRU to encode features allows for feature importance evaluations
- Transformer decoder allows for attention which reduces risk of gradient vanishing/explosion and expedites training
- Close comparison with our deep model which will use transformers



Architecture III: Deep/Foundation

DINOv2 + SHuBERT encoder, Fusion Layer, ByT5 decoder
- **DINOv2**; generate embeddings from hand/face crops [5]
- **SHuBERT**; encoder trained on ~1k hours of ASL video [1]
- **ByT5;** encoder-decoder transformer trained on ~1TB of multilingual text [6]
- Using pretrained components that have already shown promise in sentence to sentence generation will greatly improve our chance of a reasonable product for real-time. ByT5 will be finetuned by our team on How2Sign



Split Design and Leakage Prevention

- We will be using How2Sign's predefined train, validation, and test splits (Roughly 88:5:7) to ensure we have no leakage between subsets.
- How2Sign splits only include full video samples, no partials
- Within each split we will apply filters to impose a reduced vocabulary ~1000 words for one setting (Self care, fitness, food and drinks)

Note: signers are shared between splits

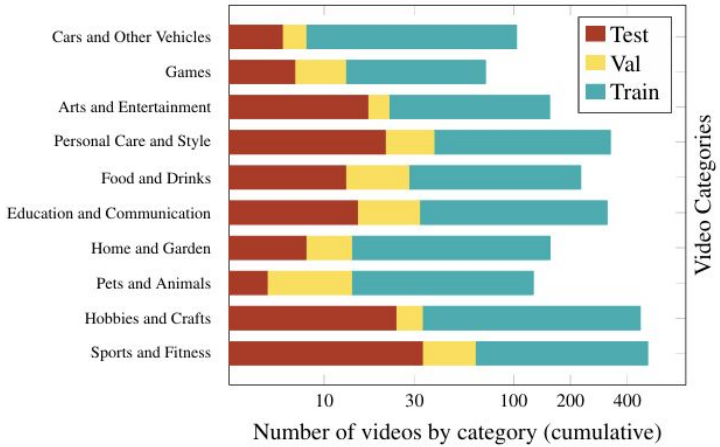


Image: How2Sign (Duarte et al., 2021) [2]

The Metrics, the Targets, and the Storyboard B Slice

Metrics: BLEURT (with BLEU) and the Confident-Error Rate

BLEURT: A metric for quantifying meaning between two sentences

Confident-Error Rate: Fraction of displayed captions (above the confidence threshold) that are incorrect

Targets: BLEURT >= 0.35 on test, confident-error rate <= 10% on the How2Sign sentences with a constrained vocabulary.

Storyboard B Slice: Test sentences with words or finger spellings that do not exist in our restricted vocabulary.

The Decision Rule and Fallback

Decision Rule:

- We will select the architecture with the highest BLEURT on the Storyboard B slice, among architectures that meet our targets

Fallback:

- If no architectures meet the decision rule, we will higher the confidence threshold. All other sentences will signify to the user that the sentence was not understood, showing the output with the confidence score.

Technical Debt

- The entire pipeline relies on media pipe which is a noted limitation of our design (entanglement)
- Our pipeline uses frozen models which may have depended on a previous media pipe version, so we will pin versions and checkpoints (external dependencies)
- We are training on offline clips, and testing on a live stream, this is why we share one preprocessing module (skew)
- We don’t collect user data as recordings are very intrusive (feedback loop)

References

[1] Gueuwou, S., Du, X., Shakhnarovich, G., Livescu, K., & Liu, A. H. (2025). SHuBERT: Self-supervised sign language representation learning via multi-stream cluster prediction. *Proceedings of ACL 2025*. https://aclanthology.org/2025.acl-long.1397/

[2] Duarte, A., Palaskar, S., Ventura, L., Ghadiyaram, D., DeHaan, K., Metze, F., Torres, J., & Giro-i-Nieto, X. (2021). How2Sign: A large-scale multimodal dataset for continuous American Sign Language. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2735–2744. https://arxiv.org/abs/2008.08143

[3] Grishchenko, I., & Bazarevsky, V. (2020, December). MediaPipe Holistic — Simultaneous face, hand and pose prediction, on device. Google AI Blog. https://research.google/blog/mediapipe-holistic-simultaneous-face-hand-and-pose-prediction-on-device/

[4] Starner, T., Weaver, J., & Pentland, A. (1998). Real-time American Sign Language recognition using desk and wearable computer based video. IEEE Transactions on Pattern Analysis and Machine Intelligence, 20(12), 1371–1375.

[5] Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al. (2024). DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research. https://arxiv.org/abs/2304.07193

[6] Xue, L., Barua, A., Constant, N., Al-Rfou, R., Narang, S., Kale, M., Roberts, A., & Raffel, C. (2022). ByT5: Towards a token-free future with pre-trained byte-to-byte models. Transactions of the Association for Computational Linguistics, 10, 291–306. https://doi.org/10.1162/tacl_a_00461